

Eric Nartey Onyame

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Education

- **Ph.D. Data Science**, University of Virginia, Virginia, USA, August 2023 - Expected May 2028. **Current GPA: 3.95/4.0**
- **M.Sc. Mathematics (Applied Mathematics)**, The University of Tennessee at Chattanooga, Tennessee, USA, August 2021 - May 2023. **GPA: 3.80/4.0**
- **B.Sc. Mathematics with Economics**, University of Cape Coast, Cape Coast, Ghana, August 2016 - July 2020. **GPA: 3.76/4.0**

Summary

Highly skilled data scientist and AI engineer with expertise in algorithm development, computational science, and applied mathematics. Experienced in developing and evaluating large language models and multimodal reasoning systems, with a proven ability to solve real-world problems. Seeking opportunities to apply my skills in a real-world setting.

Specialties

- **Trustworthy AI & Model Evaluation:** Expertise in developing fair, transparent, safe, and accountable AI systems; skilled in fine-tuning and pre-training foundational models with experience in performance evaluation, bias mitigation, and interpretability.
- **Deep Learning & Machine Learning:** Experience designing, training, and deploying supervised, unsupervised, and deep learning models using PyTorch, scikit-learn, and NumPy, including large-scale dataset processing.
- **Data Science & Statistical Analysis:** Proficient in predictive modeling, analytics, and visualization using Python, R, and Julia; strong foundation in linear algebra, probability, and statistical inference applied to real-world challenges.
- **Data Engineering & Integration:** Adept at integrating internal and external datasets via APIs, working with relational databases, and processing large-scale data for advanced analytics.
- **Software Development & Applications:** Experienced in building interactive dashboards and web applications using Python, HTML, CSS, and JavaScript to support data-driven decision-making.
- **Communication & Continuous Growth:** Effective at translating complex analytical results into actionable insights for both technical and non-technical stakeholders; committed to continuous learning in predictive modeling, large-scale data analysis, and modern AI architectures.

Professional Experience

- **PhD Researcher:** University of Virginia, Aug 2023 – Present
 - Conduct research in **Trustworthy AI**, focusing on designing and evaluating large language and multimodal models whose outputs are reliable, faithful to evidence, and worthy of user trust.
 - Develop **multilingual representation and evaluation pipelines** to improve cross-language generalization and consistency, with emphasis on high-quality reasoning in the user's native language.
 - Investigate **AI safety and monitoring**, studying unsafe or misleading model behaviors through interpretability-driven and safety-focused evaluation frameworks.
 - Explore extensions of these methods to **multimodal models**, analyzing robustness under distribution shifts and out-of-domain inputs.
 - Apply reinforcement learning techniques to improve reasoning reliability and robustness in high-stakes AI settings.
- **Researcher:** University of Tennessee at Chattanooga, Aug 2021 – May 2023
 - Applied linear and non-linear optimization to improve operational efficiency in healthcare and transportation.
 - Developed mathematical models to determine optimal locations for healthcare communication hubs, contributing to resource allocation and improved accessibility.
 - Conducted statistical and machine learning analysis of dietary data using R to model carotenoid levels and predict health conditions based on nutritional patterns.
- **Graduate Teaching Assistant,** University of Virginia, 2024 – 2025
 - Supported instruction and grading for core undergraduate and graduate courses in Data Science and AI, including:
 - * *DS 6050: Deep Learning*
 - * *DS 7800: Research Methods in Data Science*
 - * *DS 6051: Decoding Large Language Models*
 - * *DS 6600: Data Engineering I: Data Management & Visualization*
 - * *DS 1001: Foundations of Data Science*
 - Led office hours, facilitated labs, and guided hands-on projects on neural networks, optimization, data pipelines, and large language model evaluation.
 - Mentored and supported 100+ undergraduate and graduate students through project development, problem-solving, and feedback.
 - Delivered a guest lecture on real-world applications of large language models.
- **Teaching Assistant,** University of Tennessee at Chattanooga, Aug 2021 – Jun 2022
 - Assisted over 200 students in building foundational quantitative skills by delivering lectures, grading assignments, and supporting learning in statistical analysis, geometry, and algebra.

Technical Skills

- Languages: Python, R, Julia, JavaScript, SQL.
- Tools & Libraries: PyTorch, TensorFlow, PySpark, Docker, Git, NumPy, Pandas, Matplotlib, scikit-learn, LaTeX, HTML, CSS, cloud computing.

Awards and Honors

- University-wide fellowship for excellent students at the University of Virginia, February 2023.
- Research Fellowship Award for ground-breaking research in Data Science at the University of Virginia, February 2023.
- Awarded Provost scholarship to support my studies at the University of Virginia.
- Best Mathematics Graduate Student Award, University of Tennessee at Chattanooga, April 2023.
- Inductee, Pi-Mu Epsilon Mathematics Honor Society, University of Tennessee at Chattanooga, April 2022.
- Excellent Student, School of Physical Sciences, University of Cape Coast, 2017/2018, 2018/2019, 2019/2020 Academic Year Deans Award.
- Best Mathematics Student, School of Physical Sciences, University of Cape Coast, 2020.
- Best Mathematics Student, Ghanta Senior High School, 2014.

Conferences and Workshops

- 2025 Research Showcase, UVA school of Data Science
- UTC Math Workshop for High School Students, April 2023
Volunteered as a workshop facilitator, guiding high school students in exploring mathematical concepts and problem-solving techniques.
- UTC Spring Research and Arts Conference, April 2022
Presented research on “Optimization Methods for Hub Location Problems in Healthcare Systems,” showcasing innovative mathematical models and solutions for improving healthcare service delivery.
- American Mathematical Society 2022 Southeastern Sectional Meeting, October 2022
Delivered a talk on “Applications of Convex Optimization in Transportation Planning,” highlighting the practical applications and impact of optimization techniques in transportation systems.
- Workshop on Modeling Simulation of Systems and Applications (ModSim), UTC, December 2021.
- Workshop on Optimizations, Machine Learning, and Data Science Applications, UTC, May 2022.

Professional Memberships

- Member, American Mathematical Society, USA.
- Member, Charlottesville Data Science group
- **Research Committee Member**, University of Tennessee at Chattanooga (Mathematics Department), 2022-2023
Collaborate with fellow committee members to facilitate research activities, organize events, and promote research opportunities within the Mathematics Department.

Publications

- **Onyame, Eric**, Akash Ghosh, Subhadip Baidya, Sriparna Saha, Xiuying Chen, Chirag Agarwal. *CURE-Med: Curriculum-Informed Reinforcement Learning for Multilingual Medical Reasoning*. arXiv preprint arXiv:2601.13262, 2026.
Available at: <https://arxiv.org/abs/2601.13262>
(Under review)
- **Onyame, Eric Nartey**. *Covering Problem with Minimum Radius Enclosing Circle*. Master’s thesis, Department of Mathematics, University of Tennessee at Chattanooga, 2023.
Published in *Masters Theses and Doctoral Dissertations*.
Available at: <https://scholar.utc.edu/theses/796>
(This work has almost 500 downloads at the moment)

Ongoing Research Projects

- **The Robustness of Existing AI Monitors**. **Eric Onyame**, Chirag Agarwal.
(In progress)
- **Counterfactual LLM Verifiers for Math and Logic Reasoning Tasks**. Elita Lobo, **Eric Onyame**, Yair Zick, Chirag Agarwal.
(In progress)

Projects with National Impact

- **CURE-Med: Curriculum-Informed Reinforcement Learning for Multilingual Medical Reasoning**.
Lead researcher on a project addressing the reliability and safety of AI systems in multilingual healthcare settings. Designed a curriculum-informed reinforcement learning framework that improves logical correctness and language consistency of medical large language models across 13 languages, including low-resource languages. Developed *CUREMED-BENCH*, a large-scale multilingual medical reasoning benchmark to support rigorous evaluation of AI systems in high-stakes clinical and public health contexts. This work aligns with U.S. priorities in AI safety, health equity, and responsible deployment of AI in healthcare infrastructure.
- **Facility Location Optimization for Healthcare Infrastructure**.
Master’s thesis research on optimizing communication hub and facility placement using minimum enclosing circle formulations and nonlinear optimization. Developed approximation algorithms for location-allocation problems aimed at minimizing service distance and facility setup costs under demand constraints. This work contributes to efficient healthcare infrastructure planning and resource allocation, with relevance to public health systems and emergency service deployment.
- **Robustness of AI Monitoring Systems**.
Ongoing research investigating the reliability and failure modes of existing AI monitoring techniques, with emphasis on robustness under distribution shift and adversarial or unexpected model behaviors. This work addresses national priorities in AI safety and the trustworthy deployment of AI systems in high-stakes domains.

- **Counterfactual Verification for LLM Reasoning.**

Ongoing research on counterfactual-based verification methods for mathematical and logical reasoning in large language models. The project aims to improve faithfulness and correctness guarantees of AI systems, contributing to safer use of AI in decision-critical applications.